Multiple Instance Learning (MIL)

From Classical to Supervised Contrastive Deep Learning Models

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- Multiple Instance learning(MIL)
- Attention-Based Learning (Ilse et al., 2018)
- Gated Attention-Based Learning
- TransMIL (2021)
- Supervised Contrastive Learning (SC-MIL, 2023)



Overview

MVPRL

- What is MIL?
- Why is it important?
- Historical timeline (1997 → 2023)
- Core advancements: Attention, Transformers, Contrastive Learning

Motivation

- Real-world data often lacks instance-level labels.
- Example: A pathology slide labeled *“cancerous”* – but which region caused it?
- MIL addresses weak supervision : learning from “bags” of instances.

Concept: The Keychain Analogy(Dietterich et., 1997)

Introduced to explain the core logic of Multiple Instance Learning (MIL).

- Each **bag** = a **keychain**
- Each **instance** = a **key** on that chain
- If **one key opens the lock**, the **bag is positive**
- If **no key opens**, the **bag is negative**



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- **Key Insight:**
- A *positive bag* contains *at least one positive instance*,
  while a *negative bag* has *none*.

- **Real-world interpretation:**
  In drug discovery — each molecule (bag) has multiple conformations (keys).
  If any conformation binds successfully to the target protein (lock),
  the molecule is labeled *active* (positive).

Mathematical Formulation of MIL

Let $X = \{x_1, x_2, \ldots, x_K\}, Y \in \{0, 1\}$

- Each bag X contains K instances x_k.
- The bag label Y is 1 (positive) if at least one instance in the bag is positive.

$$Y = 1 \iff \exists x_k \; such\ that\ f(x_k) = 1$$

Hence, the probability of a positive bag is:

$$P(Y = 1 \mid X)$$

Classical MIL Pipeline

1. Feature Extraction

Each bag $X = \{x_1, x_2, \ldots x_K\}$

Extract instance features:

$$h_k = fo(x_k)$$

2. Aggregation

Combine instance features:

$$H = Agg(h_1, h_2, \ldots h_K)$$

Examples:

$$\max \textit{pooling: } H = \max_k h_k$$

$$\text{mean } \textit{pooling: } H = \frac{1}{K}\sum_{k=1}^{K} h_k$$

3. Classification

Predict bag label:

$$P(Y = 1 \mid X) = \sigma(w^T H + b)$$



Classical MIL Challenges

MVPRL

- Limited scalability.
- No shared feature representation.
- Non-differentiable operators (e.g., max).
- Ignored inter-instance relations.

These challenges motivated the shift toward neural MIL models that are differentiable, scalable, and capable of learning relationships among instances."



Classical MIL Limitations

MVPRL

- Hand-crafted features.
- No learned attention.
- No end-to-end optimization.
- Motivated the rise of deep learning MIL.

These rigid mathematical operations limited the flexibility and expressiveness of early MIL models, motivating the development of attention-based and deep neural MIL

Pipeline Overview

Architecture

Feature Extractor → Attention Network → Weighted Sum → Classifier

Step 1- Feature Extraction

Each instance x_k is mapped into an embedding:

$$h_k = fo(x_k)$$

Pipeline Overview

Architecture

Feature Extractor → Attention Network → Weighted Sum → Classifier

- **Step 2 – Attention Network**
- Compute attention weights for each instance:

$$a_k = \frac{exp\left(w^T tanh(Vh_k)\right)}{\sum_j exp\left(w^T tanh\left(Vh_j\right)\right)}$$

- Each $a_k \in [0, 1]$ represents the importance of instance x_k.

Pipeline Overview
Architecture

Feature Extractor → Attention Network → Weighted Sum → Classifier

- **Step 3 – Weighted Pooling**
- Aggregate features with learned weights:

$$z = \sum_k a_k \, h_k$$

- **Step 4 – Classifier**
- Predict bag-level label:

$$P(Y = 1 \mid X) = \sigma(w_c^T z + b_c)$$

Advantages of Attention MIL

- Learns instance importance.
- Differentiable and trainable.
- Robust to noise.
- Improves interpretability
- Adaptable across domains

Table 2. Results on BREAST CANCER. Experiments were run 5 times and an average (± a standard error of the mean) is reported.

METHOD	ACCURACY	PRECISION	RECALL	F-SCORE	AUC
Instance+max	0.614±0.020	0.585±0.03	0.477±0.087	0.506±0.054	0.612±0.026
Instance+mean	0.672±0.026	0.672±0.034	0.515±0.056	0.577±0.049	0.719±0.019
Embedding+max	0.607±0.015	0.558±0.013	0.546±0.070	0.543±0.042	0.650±0.013
Embedding+mean	**0.741**±0.023	**0.741**±0.023	0.654±0.054	0.689±0.034	**0.796**±0.012
Attention	**0.745**±0.018	0.718±0.021	**0.715**±0.046	**0.712**±0.025	0.775±0.016
Gated-Attention	**0.755**±0.016	**0.728**±0.016	**0.731**±0.042	**0.725**±0.023	**0.799**±0.020

Figure 5: $\phi_t(x_0)$ for a few samples $x_0 \sim p_0 = \mathcal{N}(0, 1)$

Figure 6: Probability path $p_t = \mathcal{N}(\mu t, 1)$

Limitation of Attention MIL

- **Assumes instance independence**
  Treats each instance separately without modeling relationships between them.

Fails to capture contextual or spatial dependencies
Standard attention assigns weights independently to instances, missing structural context within a bag.

While attention improves interpretability, it still simplifies relationships, motivating extensions like Gated Attention MIL and Transformer-based MIL.

Gated Attention-Based MIL

Gated Attention MIL enhances the standard attention mechanism by adding a *gating function* to improve representational power.

- **Standard Attention:**

$$\mathbf{z} = \sum_{k=1}^{K} a_k \mathbf{h}_k,$$

where:

$$a_k = \frac{\exp\{\mathbf{w}^\top \tanh\left(\mathbf{V}\mathbf{h}_k^\top\right)\}}{\sum_{j=1}^{K} \exp\{\mathbf{w}^\top \tanh\left(\mathbf{V}\mathbf{h}_j^\top\right)\}},$$

The model computes one attention score per instance using tanh as a nonlinearity.

- **Gated Attention Extension:**

$$a_k = \frac{\exp\{\mathbf{w}^\top \left(\tanh\left(\mathbf{V}\mathbf{h}_k^\top\right) \odot \mathrm{sigm}\left(\mathbf{U}\mathbf{h}_k^\top\right)\right)\}}{\sum_{j=1}^{K} \exp\{\mathbf{w}^\top \left(\tanh\left(\mathbf{V}\mathbf{h}_j^\top\right) \odot \mathrm{sigm}\left(\mathbf{U}\mathbf{h}_j^\top\right)\right)\}},$$

Adds a **sigmoid gate** σ(Uh_k) alongside the tanh(Vh_k) path.

⊙ denotes **element-wise multiplication** (the gate modulates the signal).

This combination helps the network learn **more expressive and nonlinear interactions**

Gated Attention-Based MIL

Table 13. The test AUC for MNIST-BAGS with on average 50 instances per bag for different numbers of training bags.

# of training bags	50	100	150	200	300	400	500
Instance+max	0.576 ± 0.059	0.715 ± 0.096	0.937 ± 0.045	0.992 ± 0.002	0.994 ± 0.001	0.997 ± 0.001	0.997 ± 0.001
Instance+mean	0.737 ± 0.014	0.744 ± 0.029	0.824 ± 0.012	0.813 ± 0.030	0.722 ± 0.021	0.728 ± 0.017	0.798 ± 0.011
MI-SVM	0.824 ± 0.067	0.946 ± 0.004	0.959 ± 0.002	0.967 ± 0.002	0.975 ± 0.001	0.976 ± 0.001	0.979 ± 0.001
Embedded+max	0.872 ± 0.039	0.984 ± 0.005	0.992 ± 0.001	0.996 ± 0.001	0.996 ± 0.001	0.997 ± 0.001	0.997 ± 0.001
Embedded+mean	0.841 ± 0.013	0.906 ± 0.046	0.983 ± 0.005	0.992 ± 0.001	0.996 ± 0.001	0.997 ± 0.001	0.997 ± 0.001
Attention	0.967 ± 0.010	0.982 ± 0.003	0.990 ± 0.002	0.993 ± 0.002	0.989 ± 0.003	0.994 ± 0.001	0.995 ± 0.001
Gated Attention	0.920 ± 0.042	0.977 ± 0.006	0.993 ± 0.003	0.991 ± 0.002	0.994 ± 0.002	0.995 ± 0.001	0.996 ± 0.001

This table reports the **AUC scores** for various **Multiple Instance Learning (MIL)** methods on the **MNIST-BAGS** dataset.
Each bag contains ~50 image instances, and the number of training bags varies from 50 to 500.
As the number of training bags increases, all methods show improved performance.
Traditional pooling methods (Instance + mean/max) perform poorly due to limited feature aggregation.
More advanced methods like **Embedded pooling** and **MI-SVM** perform better but still lag slightly.
Attention and **Gated Attention** networks achieve the highest and most stable AUCs (>0.99) even with fewer bags.
This shows that **attention-based MIL** learns more discriminative and robust bag representations with fewer samples.

- It's a deep learning framework that applies Transformer architectures to the Multiple Instance Learning (MIL) problem, especially for Whole Slide Image (WSI) classification in digital pathology.

TransMIL lets instances communicate before classification,
learning *not just what is present*, but *how regions relate to each other* across the slide.

Key Limitations in Classical MIL

Each patch (instance) is treated **independently**.

Attention weights depend **only** on individual patch features.

No awareness of *how* patches relate spatially or contextually.

TransMIL's Proposed Solution

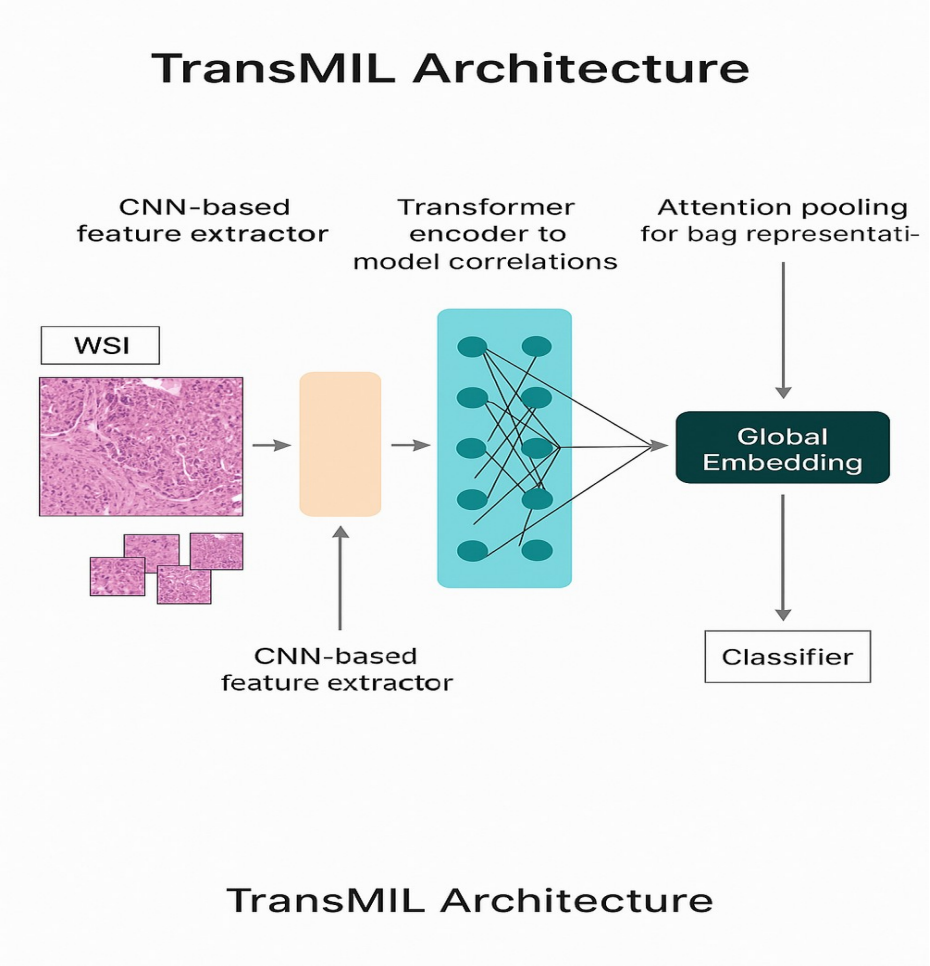
- Introduces **Transformer Encoders** with **self-attention**.
- Each patch embedding h_i interacts with *all others* before aggregation:

$$h_i = SelfAttention(h_i, h_1, h_2, ., .. h_K)$$

- Builds a context-aware representation of every patch.

TransMIL's Solution

Transformer-based MIL that models inter-patch dependencies in WSIs. Combines CNN features, positional encoding, and self-attention to capture spatial context.
Aggregates features via attention pooling for global, context-aware slide classification.



TransMIL Architecture

Core Formula

$$\mathbf{z} = \sum_{k=1}^{K} a_k \mathbf{h}_k,$$

where:

$$a_k = \frac{\exp\{\mathbf{w}^\top \tanh\left(\mathbf{V}\mathbf{h}_k^\top\right)\}}{\sum_{j=1}^{K} \exp\{\mathbf{w}^\top \tanh\left(\mathbf{V}\mathbf{h}_j^\top\right)\}},$$

$$h_i' = \mathit{TransformerEncoder}(h_i + P_i)$$

- h_i: Patch feature vector from CNN
- P_i: Positional encoding (spatial info)
- h_i': Context-enriched embedding after self-attention
- z: Bag-level representation used for classification

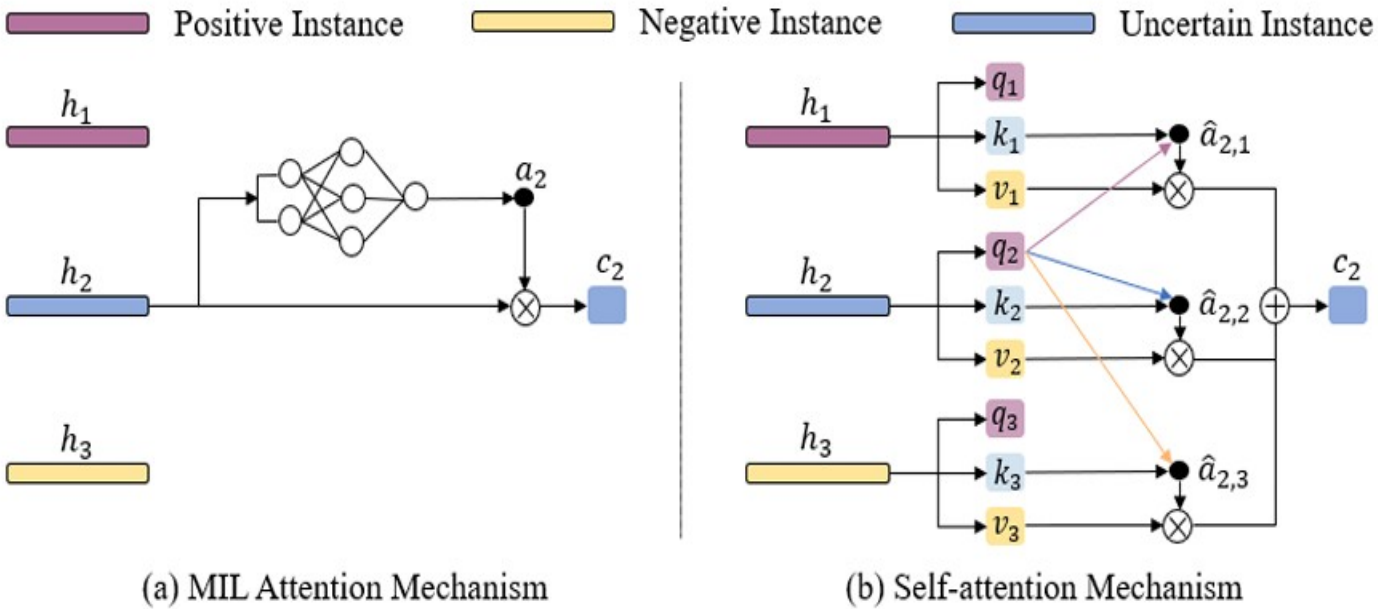
Advantages over Attention-Based MIL

- **Captures global context:** Models *all-to-all* patch interactions through self-attention.
- **Incorporates spatial correlations:** Uses 2D positional encoding to retain patch layout.
- **Learns long-range dependencies:** Enhances contextual reasoning across distant regions.
- **Improves representation power:** Builds richer, holistic bag embeddings.
- **State-of-the-art WSI performance:** Excels in pathology and other structured data tasks.
- TransMIL enhances classical attention-based MIL by **injecting spatial awareness** and **global dependencies** through Transformer encoders before pooling.
  This makes it better at capturing **structure-rich histopathology patterns** or

Limitations

- **Computationally heavy:** Self-attention scales quadratically with the number of patches.
- **Memory-intensive:** Large WSIs require patch sampling or compression.
- **Less interpretable:** Harder to visualize attention maps compared to standard MIL.
- **Needs strong regularization:** Prone to overfitting with small datasets.

TransMIL Vrs Attention MIL



(a) MIL Attention Mechanism

(b) Self-attention Mechanism

TransMIL Vs Attention MIL

The left diagram (**Attention MIL**) shows the classic attention-based Multiple Instance Learning mechanism (Ilse et al., 2018).

Each instance embedding h_i)positive, negative, or uncertain) is weighted by a learned attention score a_i ,and these weights determine their contribution to the bag representation c.
However, attention-based MIL treats each instance independently; it doesn't capture relationships or dependencies between patches.

The right diagram (**TransMIL**) introduces **self-attention**, where each instance interacts with all others through **query (q)**, **key (k)**, and **value (v)** vectors.

This mechanism models spatial and contextual correlations among instances, leading to richer and more globally aware bag embeddings.

In essence, **TransMIL** generalizes attention MIL by adding transformer-based attention, enabling the model to learn **inter-instance dependencies** crucial for complex data like whole-slide images.

Table 1: Results on CAMELYON16, TCGA-NSCLC and TCGA-RCC.

	CAMELYON16		TCGA-NSCLC		TCGA-RCC	
	Accuracy	AUC	Accuracy	AUC	Accuracy	AUC
Mean-pooling	0.6389	0.4647	0.7282	0.8401	0.9054	0.9786
Max-pooling	0.8062	0.8569	0.8593	0.9463	0.9378	0.9879
ABMIL [8]	0.8682	0.8760	0.7719	0.8656	0.8934	0.9702
PT-MTA [40]	0.8217	0.8454	0.7379	0.8299	0.9059	0.9700
MIL-RNN [23]	0.8450	0.8880	0.8619	0.9107	\	\
DSMIL [9]	0.7985	0.8179	0.8058	0.8925	0.9294	0.9841
CLAM-SB [13]	0.8760	0.8809	0.8180	0.8818	0.8816	0.9723
CLAM-MB [13]	0.8372	0.8679	0.8422	0.9377	0.8966	0.9799
TransMIL	**0.8837**	**0.9309**	**0.8835**	**0.9603**	**0.9466**	**0.9882**

This table compares **TransMIL** with several baseline MIL approaches across three pathology datasets: **CAMELYON16**, **TCGA-NSCLC**, and **TCGA-RCC**.

The metrics shown are **Accuracy** and **AUC (Area Under Curve)**, which measure classification performance and robustness respectively.

Traditional pooling methods (Mean/Max) and earlier MIL models (ABMIL, MIL-RNN, DSMIL, CLAM, PT-MTA) perform reasonably but lack spatial reasoning.
TransMIL achieves the **highest accuracy and AUC** across all datasets e.g., 0.8837 / 0.9309 on CAMELYON16 and 0.8835 / 0.9603 on TCGA-NSCLC.

This improvement comes from its **Transformer-based self-attention**, which captures **global spatial correlations** between patches.

The results confirm that modeling inter-instance dependencies enhances slide-level classification accuracy.

Overview

The problem: In pathology, with Multiple Instance Learning (MIL) applied to whole-slide images (WSIs), there's **label imbalance** at two levels:

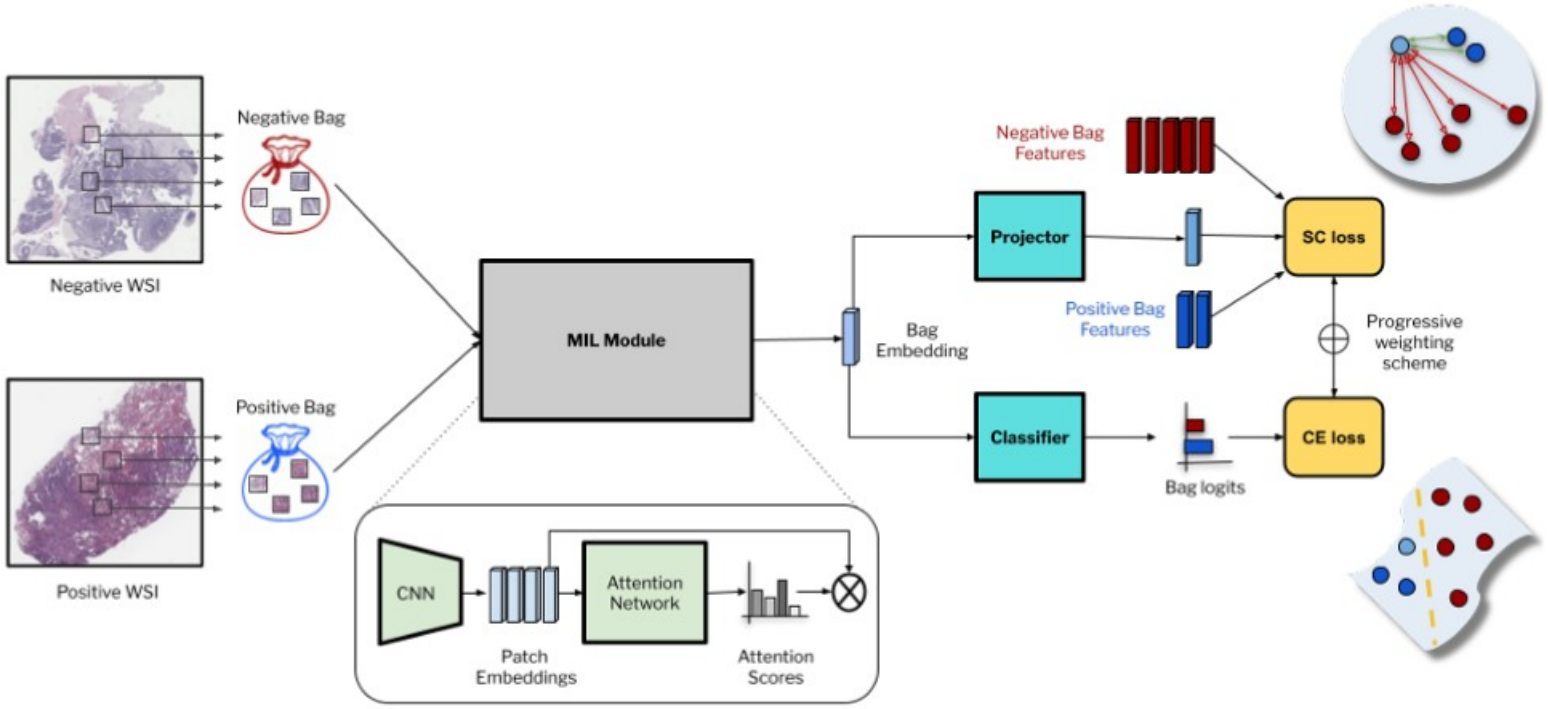
Dataset-level: Some disease classes are much rarer than others.

Slide-level (WSI-level): For a positive labelled slide, only a small subset of patches actually contribute to the positive class.

The proposed solution, SC-MIL: Integrate **supervised contrastive learning (SCL)** into an MIL framework, applied *at bag-level embeddings*, and combine with a standard classifier (cross-entropy) training.

Key insight: Decouple and gradually transition from representation (feature) learning → classifier learning helps in the imbalanced setting.

SC-MIL Architecture



SC-MIL (Supervised Contrastive Multiple Instance Learning, 2023)

•Whole-slide images (WSIs) are divided into **patches**, forming **bags** (positive or negative).
•A **CNN + Attention Network** extracts patch features and computes attention weights → bag embedding $b(X_i)$.
•The bag embedding goes to two branches:

- **Projector + SC Loss** → aligns bag features of the same class and separates different classes.
- **Classifier + CE Loss** → performs standard classification.

•A **progressive weighting scheme** combines both losses:
early training focuses on **representation learning (SC loss)**,
later training emphasizes **classification (CE loss)**.

Bag embedding via MIL

- Given a bag $X_i = \{x_{i,1}, x_{i,2}, \ldots, x_{i,n}\}$)patches from a WSI) with bag-level label Y_i.
- A feature extractor $f(\cdot)$maps patches to instance embeddings.
- An attention-module (or other aggregator) computes attention weights $\alpha_{i,j}$over instances in bag and produces bag embedding:

$$b(X_i) = \sum_{j=1}^{n} \alpha_{i,j} f(x_{i,j})$$

- This is standard attention MIL.)
- A bag-level classifier h takes b(Xi) → prediction p(Xi).

SC-MIL Methodology

Supervised Contrastive Loss at Bag Level

•After we obtain bag embedding $b(X_i)$,we apply a **projection network** $g(\cdot)$to get $z_i = g\big(b(X_i)\big)$.

•The supervised contrastive loss is:

$$\mathcal{L}_{SCL} = \sum_i -\frac{1}{|P_i^+|} \sum_{z_j \in P_i^+} \log \frac{\exp(z_i \cdot z_j/\tau)}{\sum_{z_k \in B_i} \exp(z_i \cdot z_k/\tau)}$$

- $\mathcal{P}_i^+$ =set of other bags in the batch that share the **same class** as bag i.
- $\mathcal{B}_i$ =set of all other bags in the current mini-batch.
- τ= temperature hyperparameter.

•This loss pulls together bag-embeddings of same class, pushes apart different-class ones → results in a more balanced latent space under class-imbalance.

Curriculum-style Joint Training (Feature + Classifier)

•The final training objective blends SCL + cross-entropy (CE) loss:

$$\mathcal{L}_{SC-MIL} = \beta_t \mathcal{L}_{SCL} + (1 - \beta_t) \mathcal{L}_{CE}$$

•Where; β_t decays from 1 → 0 over training iterations t. So early on, you focus more on representation (contrastive), later more on the classifier

•The authors find that **single-stage** training with the decayed blending performs better than two-stage (pure SCL first, then CE) in this MIL + imbalance setting.

Why It Works

• The contrastive loss encourages a **balanced feature space** where minority and majority classes are better separated, important for imbalanced classification. By applying contrastive learning at the bag level, the method respects the MIL structure (avoiding the assignment of patch labels incorrectly). Curriculum blending ensures good representation learning first, then robust classification learning.

• Empirically: SC-MIL outperforms baseline MIL methods + imbalance techniques on both in-distribution (ID) and out-of-distribution (OOD) datasets in pathology tasks.

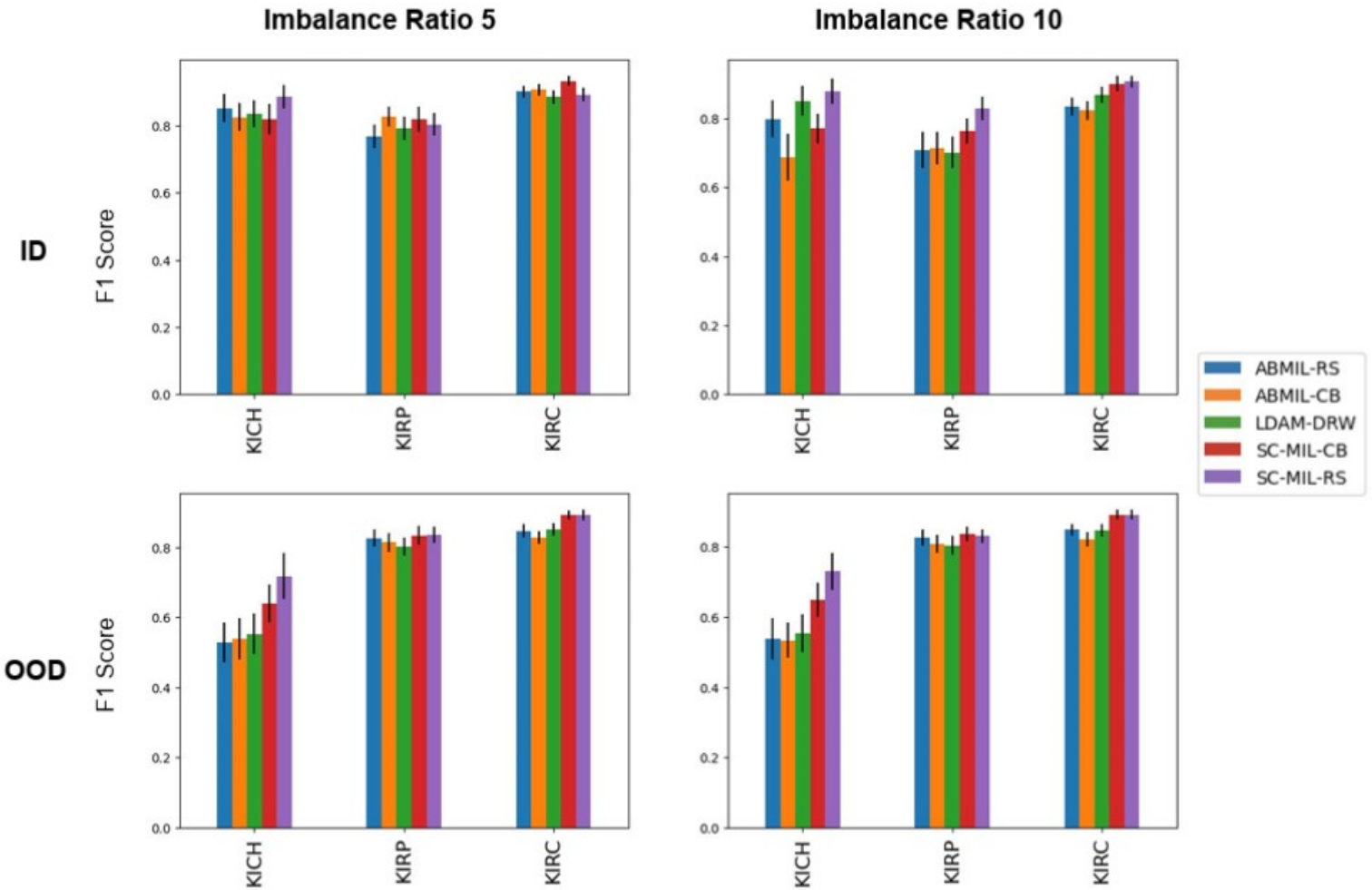
SC-MIL Comparison

Table 6. Comparison of SC-MIL with other label imbalance techniques on OOD-NSCLC test set for NSCLC subtyping (RS - Random Sampling, CB - Class Balanced)

Dataset	OOD-NSCLC					
Imbalance Ratio	1		5		10	
Metric (%)	F1	AUC	F1	AUC	F1	AUC
ABMIL-RS	71.03±5.34	92.42±2.33	19.82±7.91	75.95±4.24	12.34±6.5	88.61±2.64
ABMIL-CB	36.3±7.97	92.54±1.93	15.58±7.04	77.25±4.41	8.14±5.57	76.71±4.01
LDAM-DRW	61.63±6.81	90.31±2.5	26.28±8.11	89.57±2.56	28.85±8.02	88.36±2.89
SC-MIL-RS	**76.46±5.27**	**93.64±2.45**	37.68±7.75	91.58±2.02	**41.5±8.23**	**92.97±1.95**
SC-MIL-CB	58.82±6.68	84.83±3.08	**49.65±8.26**	**94.23±1.95**	29.06±8.2	79.04±4.4

SC-MIL achieves the best overall performance on the OOD-NSCLC dataset across all imbalance ratios.
Its supervised contrastive learning helps maintain strong representations even when minority classes are scarce.
Both Random Sampling (RS) and Class-Balanced (CB) variants outperform traditional MIL methods in F1 and AUC metrics.

SC-MIL Comparison



Impact of Sampling And Temperature

We found that SC-MIL with random sampling performs better than class-balanced sampling in most cases. We hypothesize that this is due to reduced diversity in the feature space as a side effect of oversampling the minority classes or undersampling the majority class when using class-balanced sampling, which ultimately hurts performance by interfering with feature learning

Impact of temperature. We experimented with temperature values of $\tau \in \{0.1, 0.5, 1\}$ and found that the models are generally robust to temperature changes. We reason about this through the desirable properties of representations learned through contrastive learning: uniformity in the hypersphere, inter-class separation, tolerance to potential positives, and intra-class similarity. The former is favored by low values of temperature, while higher values favor the latter. In problems with a larger number of classes, uniformity is harder to achieve, and high values of temperature. In contrast, we see that for RCC and NSCLC subtyping with 3 and 2 classes, respectively, model performance is less sensitive to changes in temperature.

The Evolution of Multiple Instance Learning (MIL)

- **MIL** redefined weakly supervised learning, labeling *bags* instead of *instances*.
- Early pooling (mean/max) offered simplicity but missed fine-grained instance insight.
- **Attention-Based MIL** introduced learnable attention, highlighting discriminative regions.
- Yet, it remained instance-isolated, ignoring relationships across the bag.
- **Gated Attention-Based MIL** deepened attention dynamics via nonlinear gating mechanisms.
- This improved interpretability and robustness, but lacked spatial awareness.
- **Transformer-based MIL (TransMIL)** bridged this gap with self-attention across patches.
- It modeled **global dependencies**, enhancing context-driven representation learning.
- **SC-MIL (2023)** unified contrastive and cross-entropy losses for balanced, generalizable learning.
- Together, these advances mark a paradigm shift, from *instance aggregation* to *relational reasoning* in deep weak supervision.

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Thank you!